

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

REGISTRATION
NUMBER

BIOLOGY

9648/03

Paper 3

17 September 2013

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Biology class, registration number and name on all the work you hand in.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The number of marks is given in brackets [] at the end of each question or part question.

Section A

Write your answers in this booklet using dark blue or black pen.

Section B

Write your answers on the separate Answer Paper using dark blue or black pen.

For Examiner's Use	
1	/ 14
2	/ 15
3	/ 11
4	/ 12
5	/ 20
TOTAL	/ 72

This document consists of **14** printed pages.

[Turn over

Section A

Answer all questions.

1. A reporter gene is a gene whose activity is relatively easy to detect. Luciferase is an enzyme naturally found in fireflies. When its substrate luciferin is present, it catalyses a chemical reaction which emits light, as shown by the equation below. Therefore, luciferase gene can be used as a reporter gene in recombinant DNA technology.

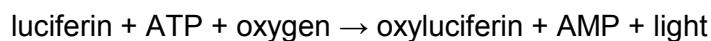


Fig. 1.1 shows a recombinant plasmid construct named “pMCS-Red Firefly Luc” used to study the regulation of *lac* operon in bacteria.

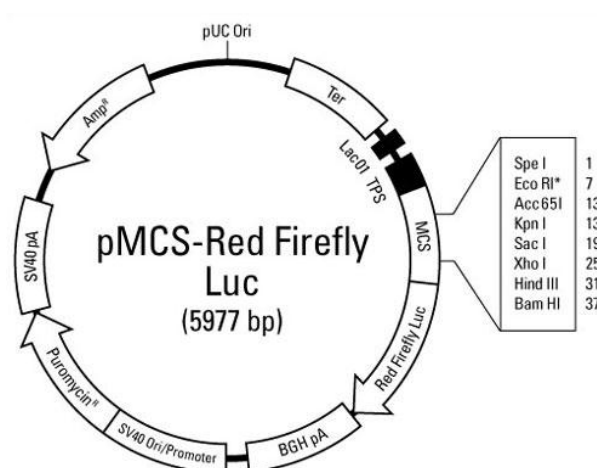


Fig. 1.1

Fig 1.2 shows the detailed structure of Red Firefly Luc insert.

Promoter of <i>lacI</i> gene	<i>lac I</i> gene	CAP binding site	Promoter	Operator	Luciferase Gene
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- (a) State the factors which you will need to consider when choosing the right restriction enzyme to construct the pMCS-Red Firefly Luc.

[2]

- (b) Both genomic and cDNA libraries of firefly are available. State where you will take the luciferase gene from for this experiment. Justify your decision.

[2]

- (c) Plasmids are introduced into bacteria via electroporation.

Outline how you can select bacterial cells transformed with recombinant plasmid containing the Red Firefly Luc insert.

[2]

A few experiments are conducted to study the regulation of *lac* operon via expression of luciferase gene in the bacterial colonies.

Complete the table below by putting in '+' (low light intensity), '++' (high light intensity) and '-' (no light emitted) in the last column.

<i>experiment</i>	<i>glucose</i>	<i>lactose</i>	<i>ampicillin</i>	<i>luciferin</i>	<i>brightness of colonies</i>
1	✓	x	✓	✓	
2	✓	✓	✓	✓	
3	x	✓	✓	✓	
4	x	✓	✓	x	
5	x	x	✓	✓	

[2]

(d) Explain the expected brightness of colonies for experiments 1 and 3.

[4]

(e) Instead of Red Firefly Luc, human insulin gene can also be cloned into pMCS plasmid shown in Fig.1.1. Suggest a suitable method to select bacterial cells transformed with recombinant plasmid containing the insulin gene insert.

[2]

[Total: 14]

2. Xeroderma pigmentosum-variant (XP-V) is an autosomal recessive disorder caused by defective DNA repair. Individuals with this disorder have sun sensitivity and increased skin cancer risk. One of the genes responsible for this disorder is the POLH gene, which encodes for DNA polymerase.

The molecular basis of XP-V was determined using DNA samples from a family in North America. Insertion of a single guanine nucleotide at position 1078 of exon 10 of the POLH gene creates a new *Bsl*I site as shown in Fig. 2.1 and is linked to an RFLP.

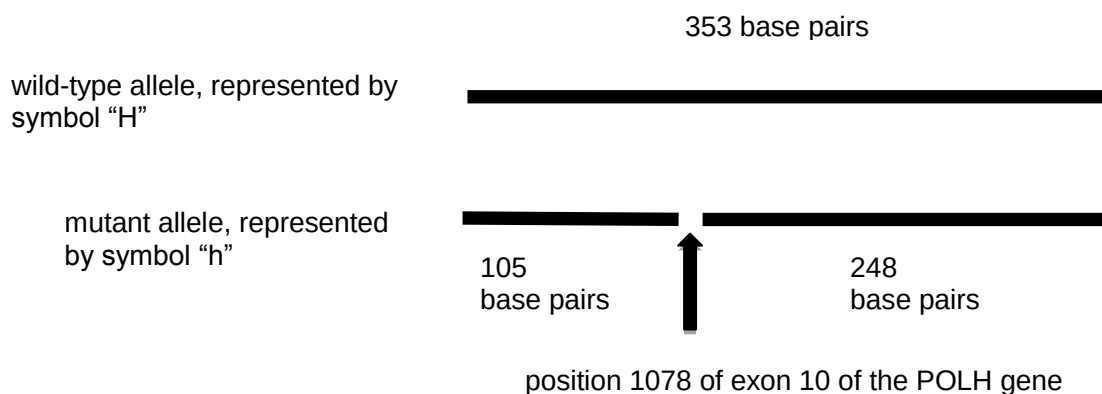


Fig. 2.1

- (a) State and explain the possible consequence of such gene mutation.

[2]

- (b) Describe two factors that are required for an RFLP marker to be used in disease detection.

[2]

(c) PCR amplification of the entire coding region of POLH gene was first performed.

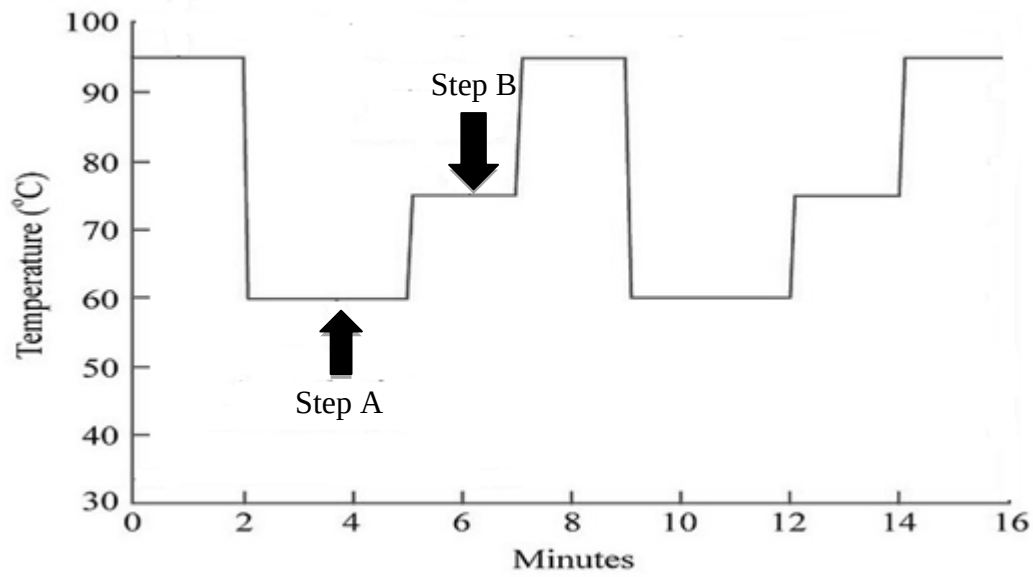


Fig. 2.2

With reference to Fig. 2.2, comment on the purpose of:

(i) Step A

[1]

(ii) Step B

[1]

Digestion of POLH gene from individuals within the family was then carried out using *Bsl*I. Fig. 2.3 shows the incomplete pedigree chart and the results obtained from gel electrophoresis.

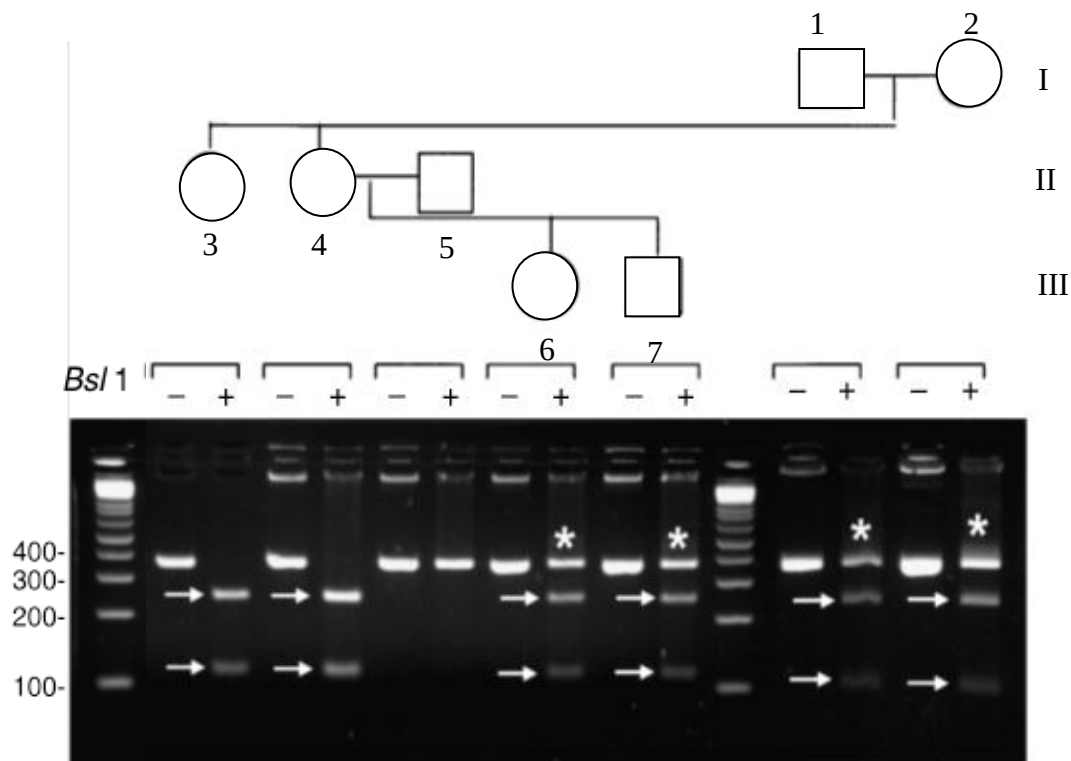


Fig. 2.3

Legend:

(-) indicates no *Bsl*I treatment; (+) indicates *Bsl*I treatment; (*) indicates normal size band; (→) indicates bands of digested DNA.

- (d) Using the information provided in the question and with reference to Fig. 2.3, state and explain the genotype of individual III-6.

[2]

(d) Outline the experimental procedure of obtaining a DNA fingerprint after the action of *Bsl*I.

[4]

(f) Explain how RFLP analysis facilitates genomic mapping.

[3]

[Total: 15]

Set 2

Name: _____ Class: _____ Registration number: _____

3. (a) Immature anther of a hibiscus flower bud was removed and the procedure shown in Fig. 3.1 was carried out to make complete plantlets.

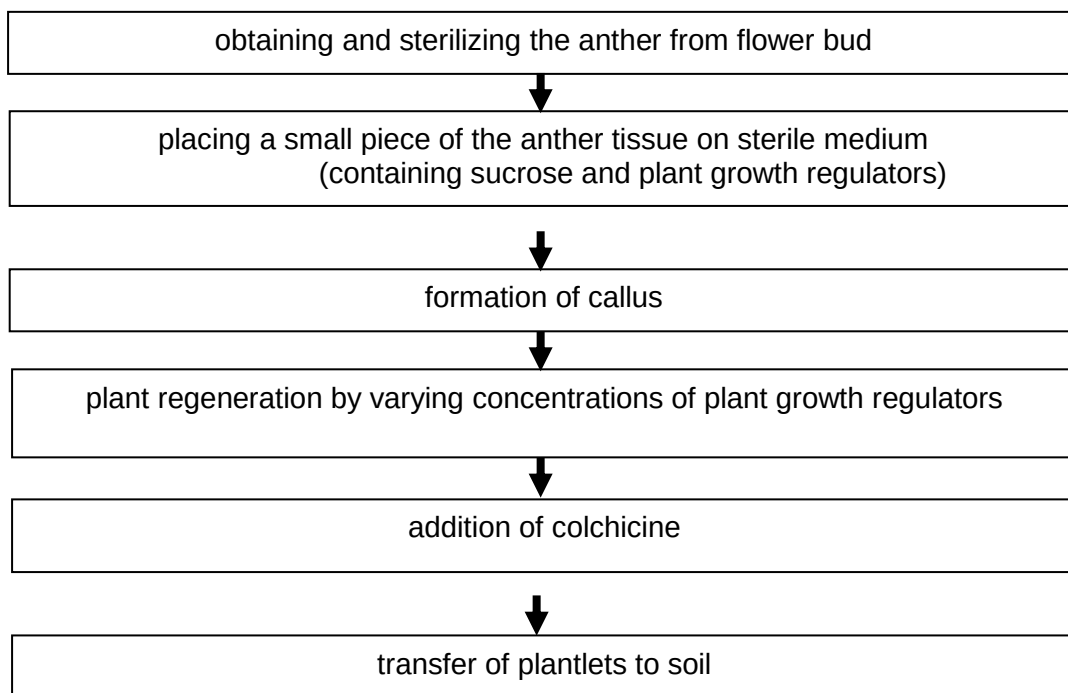


Fig. 3.1

- (i) Explain why it is important to sterilise the anther tissues.

.....
[1]
.....

- (ii) Describe briefly the importance of adding sucrose to the sterile medium.

.....
[1]
.....

(iii) Explain the purpose of adding colchicine in the final stages of the procedure.

[2]

(b) Explain the advantages of tissue culture over more traditional methods of cloning plants, such as taking cuttings or graftings.

[2]

(c) With a relevant example, explain how genetic engineering has improved the quality of crops.

[3]

(d) Researchers are experimenting with the injection of human brain cells into the brain tissues of mice. Discuss the ethical concerns that may arise.

[2]

[Total: 11]

4. Planning question

Write your answers to this question on the separate answer paper provided.

A type of genetically modified sugar cane contains higher amount of sucrose per unit mass. You are required to measure the sucrose concentration of freshly squeezed sugar cane juice from this new variant.

Your planning must be based on the assumption that you have also been provided with the following materials which you must use:

- a beaker of sugar cane juice labelled as **S**
- a 10% sucrose stock solution
- distilled water
- dilute hydrochloric acid
- sodium hydrogen carbonate solution
- thermostatically controlled water bath
- benedict's solution
- colorimeter

In addition, you may include in your plan the usage of any equipment that can be found in a standard school laboratory.

Your plan should have a clear and helpful structure to include:

- an explanation of the theory to support your practical procedure.
- a description of the method used, including the scientific reasoning behind the method
- an explanation of the dependent, independent variables and controlled variables
- how you will record your results and ensure that they are as accurate & reliable as possible.
- proposed layout of results tables and graph with clear headings and labels
- recommended safety measure.
- the correct use of technical and scientific terms.

[Total: 12]

Free Response Question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections (a), (b), etc., as indicated in the question.

- 5 (a) Explain, with examples, the treatment of genetic diseases with reference to viral and non-viral gene delivery systems. [8]
- (b) Discuss the social and ethical concerns that have arisen from gene therapy in the treatment of genetic diseases. [6]
- (c) Discuss on the different types of stem cells. [6]

--- End of Paper ---